

A New Clue to the Puzzle of the Expanding Universe

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Scientists know the universe is expanding, but they disagree about how fast. Two of the best ways to measure the speed disagree. This is the **Hubble tension**.

One way uses the **cosmic microwave background**, the faint afterglow of the Big Bang. It predicts an expansion rate of about 67. The other way measures how fast faraway galaxies move away from us, using exploding stars called **supernovae**. Hubble and James Webb **space telescope** observations give a higher value of about 73. The gap is statistically very significant.

A team of researchers wondered whether tiny **magnetic fields** from the very early universe could help. They are called primordial **magnetic fields**. The idea is that they changed how the first hydrogen formed. That would shift the moment the universe became transparent and change the "cosmic ruler" used to measure distances.

The team ran three-dimensional **simulations** and tested them against observations. Their idea stayed possible. They found a mild preference for these fields, which they say is not yet a discovery, but a meaningful hint.

Word Treasure

Hubble tension -- The disagreement between two ways of measuring how fast the universe expands.

cosmic microwave background -- The faint afterglow of the Big Bang.

supernovae -- Exploding stars.

space telescope -- A telescope that works in space to see far-away objects.

magnetic fields -- Invisible areas around magnets where magnetic forces act.

simulations -- Computer programs that copy how something works.

Break it down

Opening: the universe is expanding, but how fast?
Two ways to measure the speed and the Hubble tension
The idea of tiny magnetic fields from the early universe
Simulations, testing and what the team found

Why it matters

The puzzle suggests that scientists' best model of the universe may be missing something, so solving it could change how we understand the universe.

Test what you remember

1. What puzzle do scientists call the Hubble tension?
 - (A) Two ways of measuring how fast the universe expands disagree
 - (B) Why telescopes need special glass
 - (C) How many stars are in the Milky Way galaxy
 - (D) Why the Moon changes shape each month
2. What is the cosmic microwave background?
 - (A) The faint afterglow of the Big Bang
 - (B) A star that exploded long ago
 - (C) A kind of dust floating in space
 - (D) A new kind of space telescope
3. What value does the afterglow method predict?
 - (A) About 67
 - (B) About 7
 - (C) About 100
 - (D) About 73
4. What do the Hubble and James Webb telescopes help measure?
 - (A) How cold outer space can get
 - (B) The age of the Moon's surface
 - (C) How fast faraway galaxies move away from us
 - (D) The weather on the planet Mars
5. What did the team's idea say tiny magnetic fields might change?
 - (A) How the first hydrogen formed
 - (B) The size of planet Earth
 - (C) The color of all the stars
 - (D) The speed of light in space
6. What do the researchers say about their result?
 - (A) It has been completely ruled out
 - (B) It is a fully proven discovery
 - (C) It is a hint, not yet a discovery
 - (D) It solved the whole puzzle for good

Answer key (try the quiz first!): 1: A 2: A 3: A 4: C 5: A 6: C

MY THOUGHTS (at least 20 words):